



CENTRE FOR ADVANCED STUDIES

DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY



5-Days Online Workshop on

Artificial Intelligence: Machine Learning & Deep Learning

21 March- 25 March, 2022

sponsored by



सत्यमेव जयते

Department of Science & Technology
Government of India



Highlights



Course
Completion
Certificate



Projects,
Assignments,
Discussion



Experts led
Training



20+ hours
Online
training

Free Registration – Limited seats
Last Date of Registration: 15 March, 2022

- **Duration : 5-days**
- **Eligibility: B.Tech. 4th Year (CSE / IT /ECE)**
- **Mode: Online**

Registration Link:

<https://cas.res.in/workshops.html>



Content

- ✓ Machine Learning, Supervised and Unsupervised learning, Design and Analysis of different Machine Learning methods
- ✓ Deep Learning, Understanding of Image, Video, and Multimedia Signals
- ✓ Current Research and Role of AI, Deep Learning in Time series Signals- 1D signals
- ✓ Different CNN models and their Architecture Overview, Developed Projects demonstration and Futuristic Work, Hands-on sessions and live demonstration,
- ✓ Assignments and Discussion Sessions



Coordinator
Prof. M. K. Dutta

Director
Centre for Advanced Studies
Dr. A.P.J. Abdul Kalam Technical
University, Lucknow



Keras



TensorFlow



SPYDER



NumPy



learn



pandas



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SCHEDULE

Day	Session	
	Session 1	Session 2
Day 1	Introduction to Machine learning: Basic Paradigm, Classification and clustering, classifier surfaces, feature engineering, measuring distances between features, statistical coefficient and confusion matrix, training errors	Hands-on Session 1- Installation of programming software and IDE, Introduction to Google Colab, Basic Programming on Python, Introduction to Training resources, Import packages and loading of datasets
Day 2	Supervised Learning: Classification, Support Vector Machines (SVM), Linear and Non-linear SVMs, Kernels, Decision tree, Entropy and Information Gain, Random Forest Classifier, Naïve Bayes	Hands-on Session 2- Machine Learning, Supervised Learning, Support Vector Machines and Naïve Classifier
Day 3	Introduction to Deep Learning: Perceptron model, Activation function, Loss Function, Building Neural Networks with perceptron, Binary Cross Entropy loss, loss optimization, Gradient decent, Learning rate, Over fitting, Dropouts	Hands-on Session 3- Machine Learning Classifiers, Creating CNN models from scratch, Compiling of CNN models, Training and Testing, Plotting of curves
Day 4	Deep Learning in Computer Vision: Features Extraction, Feature Representation, Fully Connected Neural Network, Convolution on Images, Convolution Layer and Feature Maps, Pooling and Activation Function (ReLU), CNNs Applications	Hands-on Session 4- Deep learning training and architecture, feature extraction, Models training with some pertained models, Object Detection, Image Augmentation
Day 5	Deep Learning in Time Series Signals: Signals, Filtering, Augmentation, 1D Signal Processing, Features extraction, Convolution and Pooling in 1D signals, Classifications using CNN	Hands-on Session 5- Signal acquisition, signal filtering, Augmentation, Feature Extraction 1-D signals, Training and classification of 1D signals using CNN.



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